LECTURE 3

Chapter 01.02
Measuring Errors

After reading this chapter, you should be able to:

1. *find the true and relative true error,*
2. *find the approximate and relative approximate error,*
3. *relate the absolute relative approximate error to the number of significant digits at least correct in your answers, and*
4. *know the concept of significant digits.*

In any numerical analysis, errors will arise during the calculations. To be able to deal with the issue of errors, we need to

(A) identify where the error is coming from, followed by
(B) quantifying the error, and lastly
(C) minimize the error as per our needs.

In this chapter, we will concentrate on item (B), that is, how to quantify errors.

Q: What is true error?

A: True error denoted by E_t is the difference between the true value (also called the exact value) and the approximate value.

True Error = True value – Approximate value

Absolute True error = | True value − Approximate value |

Example 1

The derivative of a function $f(x)$ at a particular value of x can be approximately calculated by

$$f'(x) \approx \frac{f(x+h)-f(x)}{h}$$

of $f'(2)$ For $f(x) = 7e^{0.5x}$ and $h = 0.3$, find

a) the approximate value of $f'(2)$
b) the true value of $f'(2)$
c) the true error for part (a)

Solution

a) $$f'(x) \approx \frac{f(x+h)-f(x)}{h}$$

Definition of derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$$

Computer cannot compute $h \to 0$

$$f'(x) \approx \frac{f(x+h)-f(x)}{h}$$

put $h = 0.3$ $f'(2)$

$$f'(2) \approx \frac{f(2+0.3)-f(2)}{0.3} = 10.265$$

$f(x) = 7 \cdot e^{0.5x}$

$f'(x) = 3.5e^{0.5x}$

$f(2) = 3.5e^{1} = 9.5140$

For $x = 2$ and $h = 0.3$,

$$f'(2) \approx \frac{f(2+0.3) - f(2)}{0.3}$$
$$= \frac{f(2.3) - f(2)}{0.3}$$
$$= \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3}$$
$$= \frac{22.107 - 19.028}{0.3}$$
$$= 10.265$$

b) The exact value of $f'(2)$ can be calculated by using our knowledge of differential calculus.

$$f(x) = 7e^{0.5x}$$
$$f'(x) = 7 \times 0.5 \times e^{0.5x}$$
$$= 3.5e^{0.5x}$$

So the true value of $f'(2)$ is

$$f'(2) = 3.5e^{0.5(2)}$$
$$= 9.5140$$

c) True error is calculated as

$$E_t = \text{True value} - \text{Approximate value}$$
$$= 9.5140 - 10.265$$
$$= -0.75061$$

Absolute true error
$|E_t| = 0.75061$

The magnitude of true error does not show how bad the error is. A true error of $E_t = -0.722$ may seem to be small, but if the function given in the Example 1 were $f(x) = 7 \times 10^{-6} e^{0.5x}$, the true error in calculating $f'(2)$ with $h = 0.3$, would be $E_t = -0.75061 \times 10^{-6}$. This value of true error is smaller, even when the two problems are similar in that they use the same value of the function argument, $x = 2$ and the step size, $h = 0.3$. This brings us to the definition of relative true error.

Q: What is relative true error?
A: Relative true error is denoted by $\in_t$ and is defined as the ratio between the true error and the true value.

$$\text{Relative True Error} = \frac{\text{True Error}}{\text{True Value}}$$

Relative absolute true error
$= \left| \frac{\text{True error}}{\text{True value}} \right|$

Example 2

The derivative of a function $f(x)$ at a particular value of x can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$ and $h = 0.3$, find the relative true error at $x = 2$.

Solution

From Example 1,

$$E_t = \text{True value} - \text{Approximate value}$$
$$= 9.5140 - 10.265$$
$$= -0.75061$$

Relative true error is calculated as

$$\epsilon_t = \frac{\text{True Error}}{\text{True Value}}$$
$$= \frac{-0.75061}{9.5140}$$
$$= -0.078895$$

Absolute relative true error
$|\epsilon_t| = 0.078895$

Relative true errors are also presented as percentages. For this example,

$$\epsilon_t = -0.0758895 \times 100\%$$
$$= -7.58895\%$$

Absolute relative true errors may also need to be calculated. In such cases,

$$|\epsilon_t| = |-0.075888|$$
$$= 0.0758895$$
$$= 7.58895\%$$

Q: What is approximate error?

A: In the previous section, we discussed how to calculate true errors. Such errors are calculated only if true values are known. An example where this would be useful is when one is checking if a program is in working order and you know some examples where the true error is known. But mostly we will not have the luxury of knowing true values as why would you want to find the approximate values if you know the true values. So when we are solving a problem numerically, we will only have access to approximate values. We need to know how to quantify error for such cases.

Approximate error is denoted by E_a and is defined as the difference between the present approximation and previous approximation.

Approximate Error = Present Approximation – Previous Approximation

Absolute approximate error = |Present approximation − Previous approximation|

Example 3

The derivative of a function $f(x)$ at a particular value of x can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$h \to 0$ we get better approximation

For $f(x) = 7e^{0.5x}$ and at $x = 2$, find the following

a) $f'(2)$ using $h = 0.3$

b) $f'(2)$ using $h = 0.15$ → present approximation (better approximation)

c) approximate error for the value of $f'(2)$ for part (b)

Solution

a) The approximate expression for the derivative of a function is

$$f'(x) \approx \frac{f(x+h)-f(x)}{h}.$$

For $x = 2$ and $h = 0.3$,

$$f'(2) \approx \frac{f(2+0.3)-f(2)}{0.3}$$
$$= \frac{f(2.3)-f(2)}{0.3}$$
$$= \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3}$$
$$= \frac{22.107 - 19.028}{0.3}$$
$$= 10.265$$
$$= 10.265$$

b) Repeat the procedure of part (a) with $h = 0.15$,

$$f'(x) \approx \frac{f(x+h)-f(x)}{h}$$

For $x = 2$ and $h = 0.15$,

$$f'(2) \approx \frac{f(2+0.15)-f(2)}{0.15}$$
$$= \frac{f(2.15)-f(2)}{0.15}$$
$$= \frac{7e^{0.5(2.15)} - 7e^{0.5(2)}}{0.15}$$
$$= \frac{20.50 - 19.028}{0.15}$$
$$= 9.8799$$

Assume we don't know exact (true) value.

We compare the approximations.

c) So the approximate error, E_a is

$$\begin{aligned} E_a &= \text{Present Approximation} - \text{Previous Approximation} \\ &= 9.8799 - 10.265 \\ &= -0.38474 \end{aligned}$$

The magnitude of approximate error does not show how bad the error is . An approximate error of $E_a = -0.38300$ may seem to be small; but for $f(x) = 7\times10^{-6} e^{0.5x}$, the approximate error in calculating $f'(2)$ with $h = 0.15$ would be $E_a = -0.38474\times10^{-6}$. This value of approximate error is smaller, even when the two problems are similar in that they use the same value of the function argument, $x = 2$, and $h = 0.15$ and $h = 0.3$. This brings us to the definition of relative approximate error.

Q: What is relative approximate error?
A: Relative approximate error is denoted by $\in_a$ and is defined as the ratio between the approximate error and the present approximation.

$$\in_a = \text{Relative Approximate Error} = \frac{\text{Approximate Error}}{\text{Present Approximation}}$$

Example 4

The derivative of a function $f(x)$ at a particular value of x can be approximately calculated by

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

For $f(x) = 7e^{0.5x}$, find the relative approximate error in calculating $f'(2)$ using values from $h = 0.3$ and $h = 0.15$.

Solution

From Example 3, the approximate value of $f'(2) = 10.263$ using $h = 0.3$ and $f'(2) = 9.8800$ using $h = 0.15$.

$$\begin{aligned} E_a &= \text{Present Approximation} - \text{Previous Approximation} \\ &= 9.8799 - 10.265 \\ &= -0.38474 \end{aligned}$$

The relative approximate error is calculated as

$$\begin{aligned} \in_a &= \frac{\text{Approximate Error}}{\text{Present Approximation}} \\ &= \frac{-0.38474}{9.8799} \\ &= -0.038942 \end{aligned}$$

Relative approximate errors are also presented as percentages. For this example,

$$\epsilon_a = -0.038942 \times 100\%$$
$$= -3.8942\%$$

Absolute relative approximate errors may also need to be calculated. In this example

$$|\epsilon_a| = |-0.038942|$$
$$= 0.038942 \text{ or } 3.8942\%$$

percentage

Q: While solving a mathematical model using numerical methods, how can we use relative approximate errors to minimize the error?

A: In a numerical method that uses iterative methods, a user can calculate relative approximate error ϵ_a at the end of each iteration. The user may pre-specify a minimum acceptable tolerance called the pre-specified tolerance, ϵ_s. If the absolute relative approximate error ϵ_a is less than or equal to the pre-specified tolerance ϵ_s, that is, $|\epsilon_a| \le \epsilon_s$, then the acceptable error has been reached and no more iterations would be required.

Alternatively, one may pre-specify how many significant digits they would like to be correct in their answer. In that case, if one wants at least m significant digits to be correct in the answer, then you would need to have the absolute relative approximate error, $|\epsilon_a| \le 0.5 \times 10^{2-m}\%$.

If one wants at least m significant digits to be correct in the answer, then you would need to have the absolute approximate error,

$$|\epsilon_a| = \left|\frac{present\ value - previous\ value}{present\ value}\right| \times 100$$

$$|\epsilon_a| \le 0.5 \times 10^{2-m}\%$$

To have an approximation at

a) 2-significant digits to be correct

$m=2$ $\quad |\epsilon_a| \le 0.5 \times 10^{2-2}\% = 0.5\%$

b) 3-significant digits to be correct

$m=3$ $\quad |\epsilon_a| \le 0.5 \times 10^{2-3}\% = 0.05\%$

Example:
Find approximate value of $e^{0.7}$ at 2-significant digits correct.

Solution

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\cdots$$

Using 6 terms, we get the current approximation as

$$e^{0.7} \cong 1 + 0.7 + \frac{0.7^2}{2!} + \frac{0.7^3}{3!} + \frac{0.7^4}{4!} + \frac{0.7^5}{5!}$$
$$= 2.0136$$

Using 5 terms, we get the previous approximation as

$$e^{0.7} \cong 1 + 0.7 + \frac{0.7^2}{2!} + \frac{0.7^3}{3!} + \frac{0.7^4}{4!}$$
$$= 2.0122$$

The percentage absolute relative approximate error is

$$|\epsilon_a| = \left|\frac{2.0136 - 2.0122}{2.0136}\right| \times 100$$
$$= 0.069527\% \leq 0.5\%$$

Since $|\epsilon_a| \leq 0.5 \times 10^{2-2}\%$, at least 2 significant digits are correct in the answer of

$$e^{0.7} \cong 2.0136$$

it is not 2 significant digits.

for $m=3$ $|\epsilon_a| \leq 0.05\%$ should be but $0.069 \not\leq 0.05$

Q: But what do you mean by significant digits?

A: Significant digits are important in showing the truth one has in a reported number. For example, if someone asked me what the population of my county is, I would respond, "The population of the Hillsborough county area is 1 million". But if someone was going to give me a $100 for every citizen of the county, I would have to get an exact count. That count would have been 1,079,587 in year 2003. So you can see that in my statement that the population is 1 million, that there is only one significant digit, that is, 1, and in the statement that the population is 1,079,587, there are seven significant digits. So, how do we differentiate the number of digits correct in 1,000,000 and 1,079,587? Well for that, one may use scientific notation. For our data we show

$$1{,}000{,}000 = 1 \times 10^6$$
$$1{,}079{,}587 = 1.079587 \times 10^6$$

to signify the correct number of significant digits.

Example 5

Give some examples of showing the number of significant digits.

Solution

a) 0.0459 has three significant digits
b) 4.590 has four significant digits
c) 4008 has four significant digits
d) 4008.0 has five significant digits

Chapter 01.03
Sources of Error

After reading this chapter, you should be able to:

1. *know that there are two inherent sources of error in numerical methods – round-off and truncation error,*
2. *recognize the sources of round-off and truncation error, and*
3. *know the difference between round-off and truncation error.*

Error in solving an engineering or science problem can arise due to several factors. First, the error may be in the modeling technique. A mathematical model may be based on using assumptions that are not acceptable. For example, one may assume that the drag force on a car is proportional to the velocity of the car, but actually it is proportional to the square of the velocity of the car. This itself can create huge errors in determining the performance of the car, no matter how accurate the numerical methods you may use are. Second, errors may arise from mistakes in programs themselves or in the measurement of physical quantities. But, in applications of numerical methods itself, the two errors we need to focus on are

1. Round off error
2. Truncation error.

Q: What is round off error?

A: A computer can only represent a number approximately. For example, a number like $\frac{1}{3}$ may be represented as 0.333333 on a PC. Then the round off error in this case is $\frac{1}{3}-0.333333=0.00000033$. Then there are other numbers that cannot be represented exactly. For example, π and $\sqrt{2}$ are numbers that need to be approximated in computer calculations.

Q: What problems can be created by round off errors?
A: Twenty-eight Americans were killed on February 25, 1991. An Iraqi Scud hit the Army barracks in Dhahran, Saudi Arabia. The patriot defense system had failed to track and intercept the Scud. What was the cause for this failure?
The Patriot defense system consists of an electronic detection device called the range gate. It calculates the area in the air space where it should look for a Scud. To find out where it

should aim next, it calculates the velocity of the Scud and the last time the radar detected the Scud. Time is saved in a register that has 24 bits length. Since the internal clock of the system is measured for every one-tenth of a second, 1/10 is expressed in a 24 bit-register as 0.00011001100110011001100. However, this is not an exact representation. In fact, it would need infinite numbers of bits to represent 1/10 exactly. So, the error in the representation in decimal format is



Figure 1 Patriot missile (Courtesy of the US Armed Forces, http://www.redstone.army.mil/history/archives/patriot/patriot.html)

$$\frac{1}{10} - (0\times 2^{-1} + 0\times 2^{-2} + 0\times 2^{-3} + 1\times 2^{-4} + \ldots + 1\times 2^{-22} + 0\times 2^{-23} + 0\times 2^{-24})$$
$$= 9.537\times 10^{-8}$$

The battery was on for 100 consecutive hours, hence causing an inaccuracy of

$$= 9.537\times 10^{-8} \frac{\text{s}}{0.1\text{s}} \times 100\,\text{hr} \times \frac{3600\text{s}}{1\text{hr}}$$
$$= 0.3433\text{s}$$

The shift calculated in the range gate due to 0.3433s was calculated as 687m. For the Patriot missile defense system, the target is considered out of range if the shift was going to more than 137m.

Q: What is truncation error?
A: Truncation error is defined as the error caused by truncating a mathematical procedure. For example, the Maclaurin series for e^x is given as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots\ldots$$

This series has an infinite number of terms but when using this series to calculate e^x, only a finite number of terms can be used. For example, if one uses three terms to calculate e^x, then

$$e^x \approx 1 + x + \frac{x^2}{2!}.$$

the truncation error for such an approximation is

$$\text{Truncation error} = e^x - \left(1 + x + \frac{x^2}{2!}\right),$$

$$= \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots\ldots\ldots\ldots$$

But, how can truncation error be controlled in this example? We can use the concept of relative approximate error to see how many terms need to be considered. Assume that one is calculating $e^{1.2}$ using the Maclaurin series, then

$$e^{1.2} = 1 + 1.2 + \frac{1.2^2}{2!} + \frac{1.2^3}{3!} + \ldots\ldots\ldots\ldots$$

Let us assume one wants the absolute relative approximate error to be less than 1%. In Table 1, we show the value of $e^{1.2}$, approximate error and absolute relative approximate error as a function of the number of terms, n.

n	$e^{1.2}$	E_a	$\lvert\epsilon_a\rvert\%$
1	1	-	-
2	2.2	1.2	54.546
3	2.92	0.72	24.658
4	3.208	0.288	8.9776
5	3.2944	0.0864	2.6226
6	3.3151	0.020736	0.62550

Using 6 terms of the series yields a $\lvert\epsilon_a\rvert < 1\%$.

Q: Can you give me other examples of truncation error?

A: In many textbooks, the Maclaurin series is used as an example to illustrate truncation error. This may lead you to believe that truncation errors are just chopping a part of the series. However, truncation error can take place in other mathematical procedures as well. For example to find the derivative of a function, we define

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

But since we cannot use $\Delta x \to 0$, we have to use a finite value of Δx, to give

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

So the truncation error is caused by choosing a finite value of Δx as opposed to a $\Delta x \to 0$.

For example, in finding $f'(3)$ for $f(x) = x^2$, we have the exact value calculated as follows.

$$f(x) = x^2$$

From the definition of the derivative of a function,

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{(x + \Delta x)^2 - (x)^2}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{x^2 + 2x\Delta x + (\Delta x)^2 - x^2}{\Delta x}$$
$$= \lim_{\Delta x \to 0} (2x + \Delta x)$$
$$= 2x$$

This is the same expression you would have obtained by directly using the formula from your differential calculus class

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

By this formula for

$$f(x) = x^2$$
$$f'(x) = 2x$$

The exact value of $f'(3)$ is

$$f'(3) = 2 \times 3$$
$$= 6$$

If we now choose $\Delta x = 0.2$, we get

$$f'(3) \approx \frac{f(3+0.2) - f(3)}{0.2}$$
$$= \frac{f(3.2) - f(3)}{0.2}$$
$$= \frac{3.2^2 - 3^2}{0.2}$$
$$= \frac{10.24 - 9}{0.2}$$
$$= \frac{1.24}{0.2}$$
$$= 6.2$$

We purposefully chose a simple function $f(x) = x^2$ with value of $x = 2$ and $\Delta x = 0.2$ because we wanted to have no round-off error in our calculations so that the truncation error can be isolated. The truncation error in this example is

$$6 - 6.2 = -0.2.$$

Can you reduce the truncate error by choosing a smaller Δx?

Another example of truncation error is the numerical integration of a function.

$$I = \int_a^b f(x)dx$$

Exact calculations require us to calculate the area under the curve by adding the area of the rectangles as shown in Figure 2. However, exact calculations requires an infinite number of such rectangles. Since we cannot choose an infinite number of rectangles, we will have truncation error.

For example, to find

$$\int_3^9 x^2 dx,$$

we have the exact value as

$$\int_3^9 x^2 dx = \left[\frac{x^3}{3}\right]_3^9$$
$$= \left[\frac{9^3 - 3^3}{3}\right]$$
$$= 234$$

If we now choose to use two rectangles of equal width to approximate the area (see Figure 2) under the curve, the approximate value of the integral

$$\int_3^9 x^2 dx = (x^2)\Big|_{x=3}(6-3) + (x^2)\Big|_{x=6}(9-6)$$
$$= (3^2)3 + (6^2)3$$
$$= 27 + 108$$
$$= 135$$

$\int_a^b f(x)\,dx$ = sum of the area under $f(x)$ over x-axis on $[a,b]$, $f \geq 0$ on $[a,b]$

$= 135$

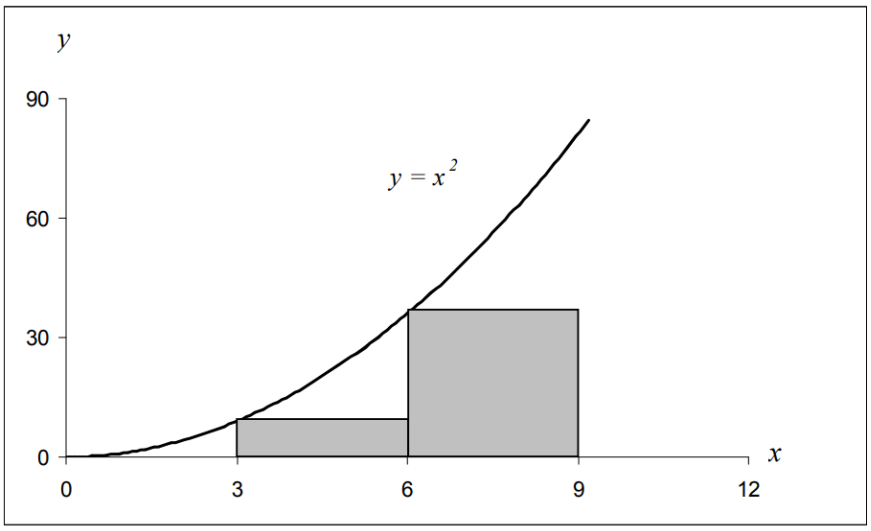


Figure 2 Plot of $y = x^2$ showing the approximate area under the curve from $x = 3$ to $x = 9$ using two rectangles.

Again, we purposefully chose a simple example because we wanted to have no round off error in our calculations. This makes the obtained error purely truncation. The truncation error is

$$234 - 135 = 99$$

Can you reduce the truncation error by choosing more rectangles as given in Figure 3? What is the truncation error?

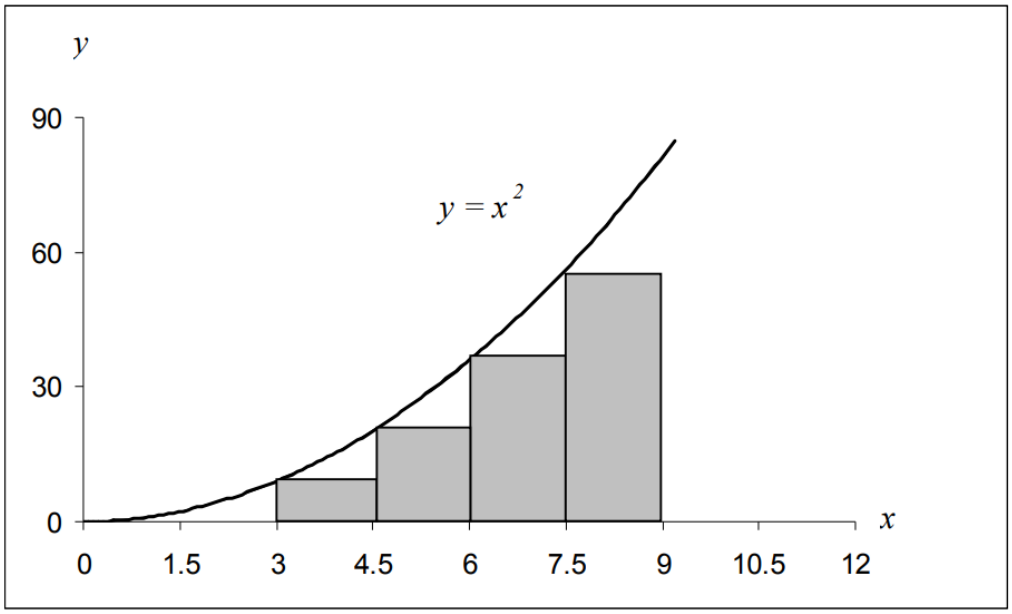


Figure 3 Plot of $y = x^2$ showing the approximate area under the curve from $x = 3$ to $x = 9$ using four rectangles.

The Following lecture note is from the book of Chapra:

4.1 ERRORS

Engineers and scientists constantly find themselves having to accomplish objectives based on uncertain information. Although perfection is a laudable goal, it is rarely if ever attained. For example, despite the fact that the model developed from Newton's second law is an excellent approximation, it would never in practice exactly predict the jumper's fall. A variety of factors such as winds and slight variations in air resistance would result in deviations from the prediction. If these deviations are systematically high or low, then we might need to develop a new model. However, if they are randomly distributed and tightly grouped around the prediction, then the deviations might be considered negligible and the model deemed adequate. Numerical approximations also introduce similar discrepancies into the analysis.

4.1.1 Accuracy and Precision

The errors associated with both calculations and measurements can be characterized with regard to their accuracy and precision. *Accuracy* refers to how closely a computed or measured value agrees with the true value. *Precision* refers to how closely individual computed or measured values agree with each other.

These concepts can be illustrated graphically using an analogy from target practice. The bullet holes on each target in Fig. 4.1 can be thought of as the predictions of a numerical technique, whereas the bull's-eye represents the truth. *Inaccuracy* (also called *bias*) is defined as systematic deviation from the truth. Thus, although the shots in Fig. 4.1*c* are more tightly grouped than in Fig. 4.1*a*, the two cases are equally biased because they are both centered on the upper left quadrant of the target. *Imprecision* (also called *uncertainty*), on the other hand, refers to the magnitude of the scatter. Therefore, although Fig. 4.1*b* and *d* are equally accurate (i.e., centered on the bull's-eye), the latter is more precise because the shots are tightly grouped.

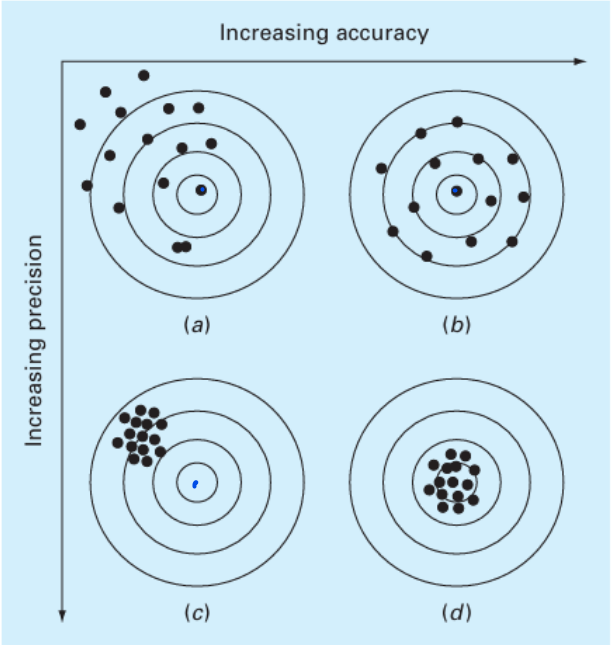


FIGURE 4.1
An example from marksmanship illustrating the concepts of accuracy and precision: (*a*) inaccurate and imprecise, (*b*) accurate and imprecise, (*c*) inaccurate and precise, and (*d*) accurate and precise.

4.1.2 Error Definitions

Numerical errors arise from the use of approximations to represent exact mathematical operations and quantities. For such errors, the relationship between the exact, or true, result and the approximation can be formulated as

$$\text{True value} = \text{approximation} + \text{error} \tag{4.1}$$

By rearranging Eq. (4.1), we find that the numerical error is equal to the discrepancy between the truth and the approximation, as in

$$E_t = \text{true value} - \text{approximation} \tag{4.2}$$

where E_t is used to designate the exact value of the error. The subscript t is included to designate that this is the "true" error. This is in contrast to other cases, as described shortly, where an "approximate" estimate of the error must be employed. Note that the true error is commonly expressed as an absolute value and referred to as the *absolute error*.

A shortcoming of this definition is that it takes no account of the order of magnitude of the value under examination. For example, an error of a centimeter is much more significant

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$$\text{True fractional relative error} = \frac{\text{true value} - \text{approximation}}{\text{true value}}$$

The relative error can also be multiplied by 100% to express it as

$$\varepsilon_t = \frac{\text{true value} - \text{approximation}}{\text{true value}} 100\% \tag{4.3}$$

where ε_t designates the true percent relative error.

Relative approximate error is:

$$\varepsilon_a = \frac{\text{present approximation} - \text{previous approximation}}{\text{present approximation}} 100\% \tag{4.5}$$

which can also lead to a negative error. Often, when performing computations, we may not be concerned with the sign of the error but are interested in whether the absolute value of the percent relative error is lower than a prespecified tolerance ε_s. Therefore, it is often useful to employ the absolute value of Eq. (4.5). For such cases, the computation is repeated until

$$|\varepsilon_a| < \varepsilon_s \tag{4.6}$$

This relationship is referred to as a *stopping criterion*. If it is satisfied, our result is assumed to be within the prespecified acceptable level ε_s. Note that for the remainder of this text, we almost always employ absolute values when using relative errors.

It is also convenient to relate these errors to the number of significant figures in the approximation. It can be shown (Scarborough, 1966) that if the following criterion is met, we can be assured that the result is correct to *at least n* significant figures.

$$\varepsilon_s = (0.5 \times 10^{2-n})\% \tag{4.7}$$

EXAMPLE 4.1 Error Estimates for Iterative Methods

Problem Statement. In mathematics, functions can often be represented by infinite series. For example, the exponential function can be computed using

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} \tag{E4.1.1}$$

Thus, as more terms are added in sequence, the approximation becomes a better and better estimate of the true value of e^x. Equation (E4.1.1) is called a *Maclaurin series expansion*.

Starting with the simplest version, $e^x = 1$, add terms one at a time in order to estimate $e^{0.5}$. After each new term is added, compute the true and approximate percent relative errors with Eqs. (4.3) and (4.5), respectively. Note that the true value is $e^{0.5} = 1.648721\ldots$. Add terms until the absolute value of the approximate error estimate ε_a falls below a prespecified error criterion ε_s conforming to three significant figures.

Solution. First, Eq. (4.7) can be employed to determine the error criterion that ensures a result that is correct to at least three significant figures:

$$\varepsilon_s = (0.5 \times 10^{2-3})\% = 0.05\%$$

Thus, we will add terms to the series until ε_a falls below this level.

The first estimate is simply equal to Eq. (E4.1.1) with a single term. Thus, the first estimate is equal to 1. The second estimate is then generated by adding the second term as in

$$e^x = 1 + x$$

or for $x = 0.5$

$$e^{0.5} = 1 + 0.5 = 1.5$$

This represents a true percent relative error of [Eq. (4.3)]

$$\varepsilon_t = \left|\frac{1.648721 - 1.5}{1.648721}\right| \times 100\% = 9.02\%$$

Equation (4.5) can be used to determine an approximate estimate of the error, as in

$$\varepsilon_a = \left|\frac{1.5 - 1}{1.5}\right| \times 100\% = 33.3\%$$

Because ε_a is not less than the required value of ε_s, we would continue the computation by adding another term, $x^2/2!$, and repeating the error calculations. The process is continued until $|\varepsilon_a| < \varepsilon_s$. The entire computation can be summarized as

Terms	Result	ε_t, %	ε_a, %
1	1	39.3	
2	1.5	9.02	33.3
3	1.625	1.44	7.69
4	1.645833333	0.175	1.27
5	1.648437500	0.0172	0.158
6	1.648697917	0.00142	0.0158

Thus, after six terms are included, the approximate error falls below $\varepsilon_s = 0.05\%$, and the computation is terminated. However, notice that, rather than three significant figures, the result is accurate to five! This is because, for this case, both Eqs. (4.5) and (4.7) are conservative. That is, they ensure that the result is at least as good as they specify. Although, this is not always the case for Eq. (4.5), it is true most of the time.

The following part is from Burden's book:

1.2 Round-off Errors and Computer Arithmetic

The arithmetic performed by a calculator or computer is different from the arithmetic in algebra and calculus courses. You would likely expect that we always have as true statements things such as $2+2 = 4$, $4 \cdot 8 = 32$, and $(\sqrt{3})^2 = 3$. However, with *computer* arithmetic we expect exact results for $2+2 = 4$ and $4 \cdot 8 = 32$, but we will not have precisely $(\sqrt{3})^2 = 3$. To understand why this is true we must explore the world of finite-digit arithmetic.

In our traditional mathematical world we permit numbers with an infinite number of digits. The arithmetic we use in this world *defines* $\sqrt{3}$ as that unique positive number that when multiplied by itself produces the integer 3. In the computational world, however, each representable number has only a fixed and finite number of digits. This means, for example, that only rational numbers—and not even all of these—can be represented exactly. Since $\sqrt{3}$ is not rational, it is given an approximate representation, one whose square will not be precisely 3, although it will likely be sufficiently close to 3 to be acceptable in most situations. In most cases, then, this machine arithmetic is satisfactory and passes without notice or concern, but at times problems arise because of this discrepancy.

Error due to rounding should be expected whenever computations are performed using numbers that are not powers of 2. Keeping this error under control is extremely important when the number of calculations is large.

The error that is produced when a calculator or computer is used to perform real-number calculations is called **round-off error**. It occurs because the arithmetic performed in a machine involves numbers with only a finite number of digits, with the result that calculations are performed with only approximate representations of the actual numbers. In a computer, only a relatively small subset of the real number system is used for the representation of all the real numbers. This subset contains only rational numbers, both positive and negative, and stores the fractional part, together with an exponential part.

1. Compute the absolute error and relative error in approximations of p by p^*.

a. $p = \pi, p^* = 22/7$
b. $p = \pi, p^* = 3.1416$
c. $p = e, p^* = 2.718$
d. $p = \sqrt{2}, p^* = 1.414$
e. $p = e^{10}, p^* = 22000$
f. $p = 10^{\pi}, p^* = 1400$
g. $p = 8!, p^* = 39900$
h. $p = 9!, p^* = \sqrt{18\pi}(9/e)^9$

2. Find the largest interval in which p^* must lie to approximate p with relative error at most 10^{-4} for each value of p.

a. π
b. e
c. $\sqrt{2}$
d. $\sqrt[3]{7}$

3. Suppose p^* must approximate p with relative error at most 10^{-3}. Find the largest interval in which p^* must lie for each value of p.

a. 150
b. 900
c. 1500
d. 90

4. Perform the following computations (i) exactly, (ii) using three-digit chopping arithmetic, and (iii) using three-digit rounding arithmetic. (iv) Compute the relative errors in parts (ii) and (iii).

a. $\frac{4}{5} + \frac{1}{3}$
b. $\frac{4}{5} \cdot \frac{1}{3}$
c. $\left(\frac{1}{3} - \frac{3}{11}\right) + \frac{3}{20}$
d. $\left(\frac{1}{3} + \frac{3}{11}\right) - \frac{3}{20}$

5. Use three-digit rounding arithmetic to perform the following calculations. Compute the absolute error and relative error with the exact value determined to at least five digits.

a. $133 + 0.921$
b. $133 - 0.499$
c. $(121 - 0.327) - 119$
d. $(121 - 119) - 0.327$
e. $\dfrac{\frac{13}{14} - \frac{6}{7}}{2e - 5.4}$
f. $-10\pi + 6e - \frac{3}{62}$
g. $\left(\frac{2}{9}\right) \cdot \left(\frac{9}{7}\right)$
h. $\dfrac{\pi - \frac{22}{7}}{\frac{1}{17}}$

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1.3 ERRORS AND THEIR COMPUTATIONS

There are two kinds of numbers, *exact* and *approximate* numbers. Examples of exact numbers are 1, 2, 3, …, 1/2, 3/2, …, $\sqrt{2}, \pi$, e, etc., written in this manner. Approximate numbers are those that represent the numbers to a certain degree of accuracy. Thus, an approximate value of π is 3.1416, or if we desire a better approximation, it is 3.14159265. But we cannot write the *exact* value of π.

The digits that are used to express a number are called *significant digits* or *significant figures*. Thus, the numbers 3.1416, 0.66667 and 4.0687 contain five significant digits each. The number 0.00023 has, however, only two significant digits, viz., 2 and 3, since the zeros serve only to fix the position of the decimal point. Similarly, the numbers 0.00145, 0.000145 and 0.0000145 all have three significant digits. In case of ambiguity, the scientific notation should be used. For example, in the number 25,600, the number of significant figures is uncertain, whereas the numbers 2.56×10^4, 2.560×10^4 and 2.5600×10^4 have three, four and five significant digits, respectively.